# Assignment approach and decision log

## Input and inspection

The supplied update file supersedes the original Markdown wording: the source is the PDF. The bundle contains ct200_manual.pdf (6 pages) and ct200_manual_v2.pdf (7 pages). Both expose extractable text in Drive's PDF viewer; this implementation independently checks each page's text layer and only invokes OCR on an empty page.

Notable irregularities observed:

- Heading 2.1.1.1 Battery Life Under Typical Use skips the expected 2.1.1 level.
- Page 3 presents 3.4 Auto Shutoff before 3.3 Result Display and Classification. Preserve original reading order; do not sort by number.
- The blood-pressure classification list is numbered like sections but is body content, not section hierarchy.
- Specifications and error-code tables can be emitted as flattened text by PDF layout engines. Attempt table detection and preserve text fallback with page provenance.
- v2 adds a page and changes values/wording. A filename or page-count change must not be mistaken for a single changed section.
- High-impact observed v1 to v2 edits include battery cycles 300 to 250 and low-battery threshold 15% to 10% in 2.1.1.1, cuff inflation increments 40 to 30 mmHg in 3.2, an added E6 error code reflected by 4.2/4.3, and new section 5.3 Data Export. These values should be checked against the rendered PDF before anyone uses a generated test idea.

## Extraction and tree

Pypdf extracts positioned text fragments and font metadata. Fragments are grouped by baseline and sorted into reading order. Numbered section headings are recognized using section-number syntax plus layout/font cues; the list-number false-positive is tested. A map of existing numeric prefixes determines parentage. When a number jumps levels, parent to the nearest existing prefix and retain the full number and reported level. Paragraphs/lists/tables are content children of the active heading. All nodes retain page provenance and a canonical content hash.

When a page contains no text layer, attempt OCRmyPDF (which uses Tesseract/Ghostscript) if available. If OCR is unavailable or errors, preserve a warning and an empty-page record; do not invent text or mark extraction complete without warning. Borderless tables are reconstructed only when explicit column headers are found; output includes raw rows and an alignment warning. Images are recorded as references, not clinically interpreted. pypdf is not itself OCR software and PDF text extraction cannot infer semantic structure; see the [official extraction notes](https://pypdf.readthedocs.io/en/stable/user/extract-text.html).

## Version matching and traceability

The primary matching key is the explicit section number. Unnumbered title/heading nodes use normalized heading path and parent path; the fallback is heuristic and returned as such. For a logical match, compare the canonical subtree hash. A changed hash marks any prior generation tied to that node and version stale. Parent sections also report a subtree change when descendants change. If a node is absent or the match is ambiguous, report unknown rather than claiming current. Selection records pin the exact version, node keys, and hashes used to create the prompt. Retrieval compares those pins to the requested/current version and exposes changed/missing sources.

This approach favors an explainable, auditable diff for this short sample over semantic embeddings. Semantic matching could improve renamed sections but adds false-match risk; it is not required to demonstrate stale traceability.

## LLM output and failure policy

The prompt requests 3-5 test-case ideas as JSON. Pydantic validates IDs, source references, rationale, steps, and expected result. Invalid output gets one repair attempt when using a live provider; if still invalid, return an error and store no successful generation. A network/provider failure is surfaced, not converted to made-up output. If no API key is configured, the clearly labelled offline demo generator supports local end-to-end testing. Duplicate requests are allowed as separate generation records so retries remain auditable; no hidden deduplication.

Generated test ideas are not approved tests, clinical advice, or a substitute for engineering review. The stored output records the selection ID, exact source version/hash, model/provider, and prompt hash.

## Time / quality tradeoffs

Priority was given to source provenance, stable section keys, explicit hashes, v1/v2 diffing, stale retrieval, and runnable tests. No frontend, authentication, auto-regeneration, or generalized document platform was added because the brief excludes or deprioritizes them. The parser is deliberately conservative: ambiguous content remains text with warnings instead of being silently reclassified.

## Limitations

OCR, table reconstruction, semantic heading recognition, and path matching need human review. A number-looking list item can still confuse a heuristic in a document with unusual typography. Hashes are sensitive to extraction whitespace; normalization reduces harmless differences but can’t prove semantic equivalence. This prototype has not been validated for regulated use.